ENERGY
Learning Hub

RAG-Powered Knowledge System for Oil, Gas & Energy Education

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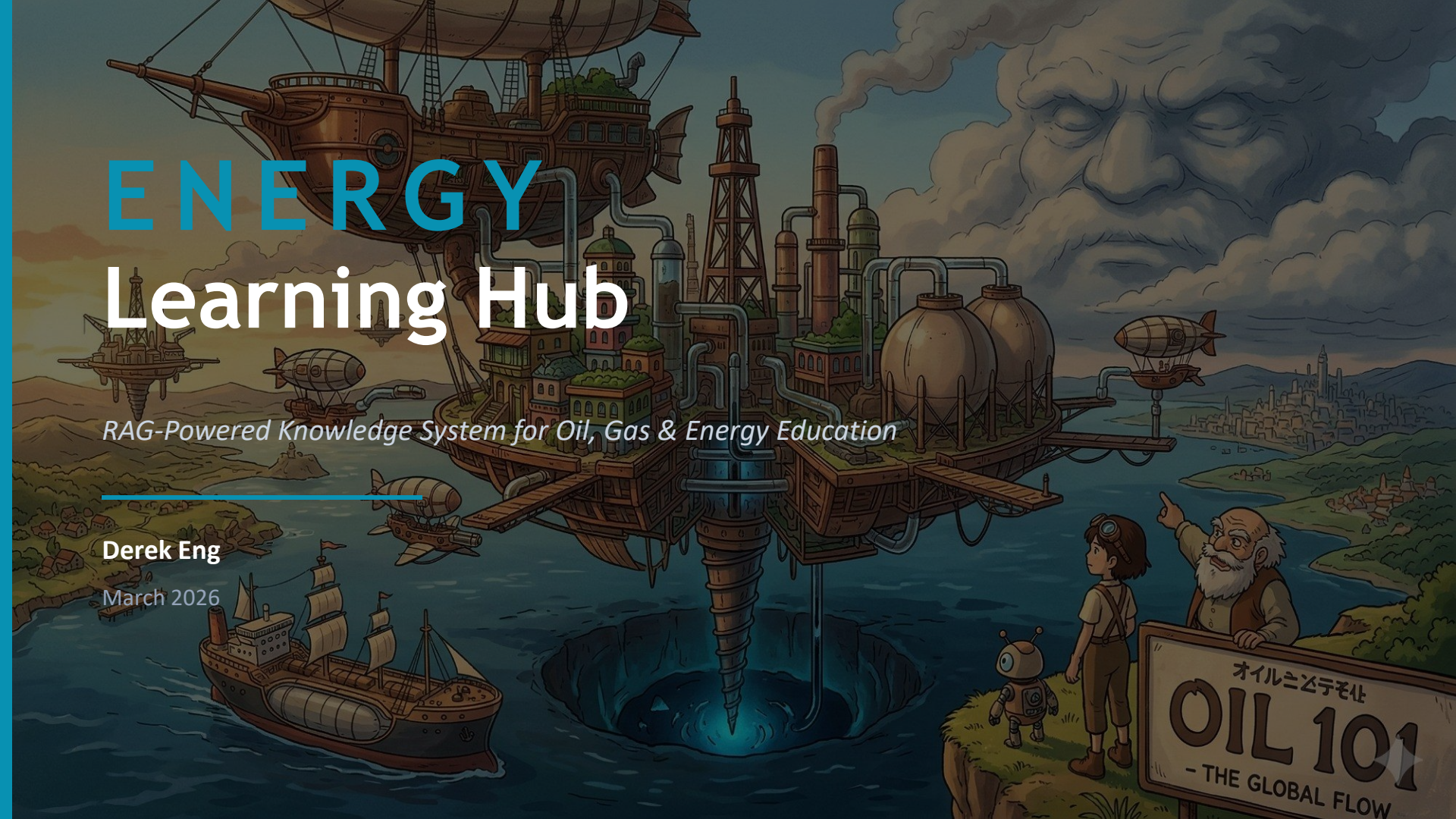


The Challenge

Why energy newcomers need a smarter way to learn

Fragmented Knowledge

Energy topics span geology, economics, policy, and engineering — scattered across dense textbooks and reports

Steep Learning Curve

Newcomers face jargon-heavy material with no guided pathway from basics to advanced concepts

No Interactive Access

Static PDFs and textbooks don’t answer follow-up questions or adapt to a learner’s level

Target Audience

Who the Energy Learning Hub is built for

1

New Industry Entrants

Graduates and career-switchers entering oil, gas, or renewable energy sectors for the first time

2

English-Speaking Learners

Professionals and students comfortable with English-language technical content and terminology

3

Self-Directed Explorers

Curious individuals who want guided, interactive access to expert-level energy knowledge at their own pace

The Solution

An AI-powered, interactive learning platform grounded in authoritative energy texts

RAG-Powered Answers

Responses grounded in curated PDFs with page-level citations

LangGraph Router

Gemini and Qwen orchestrated via LangGraph for resilience

Tiered FAQ Pathways

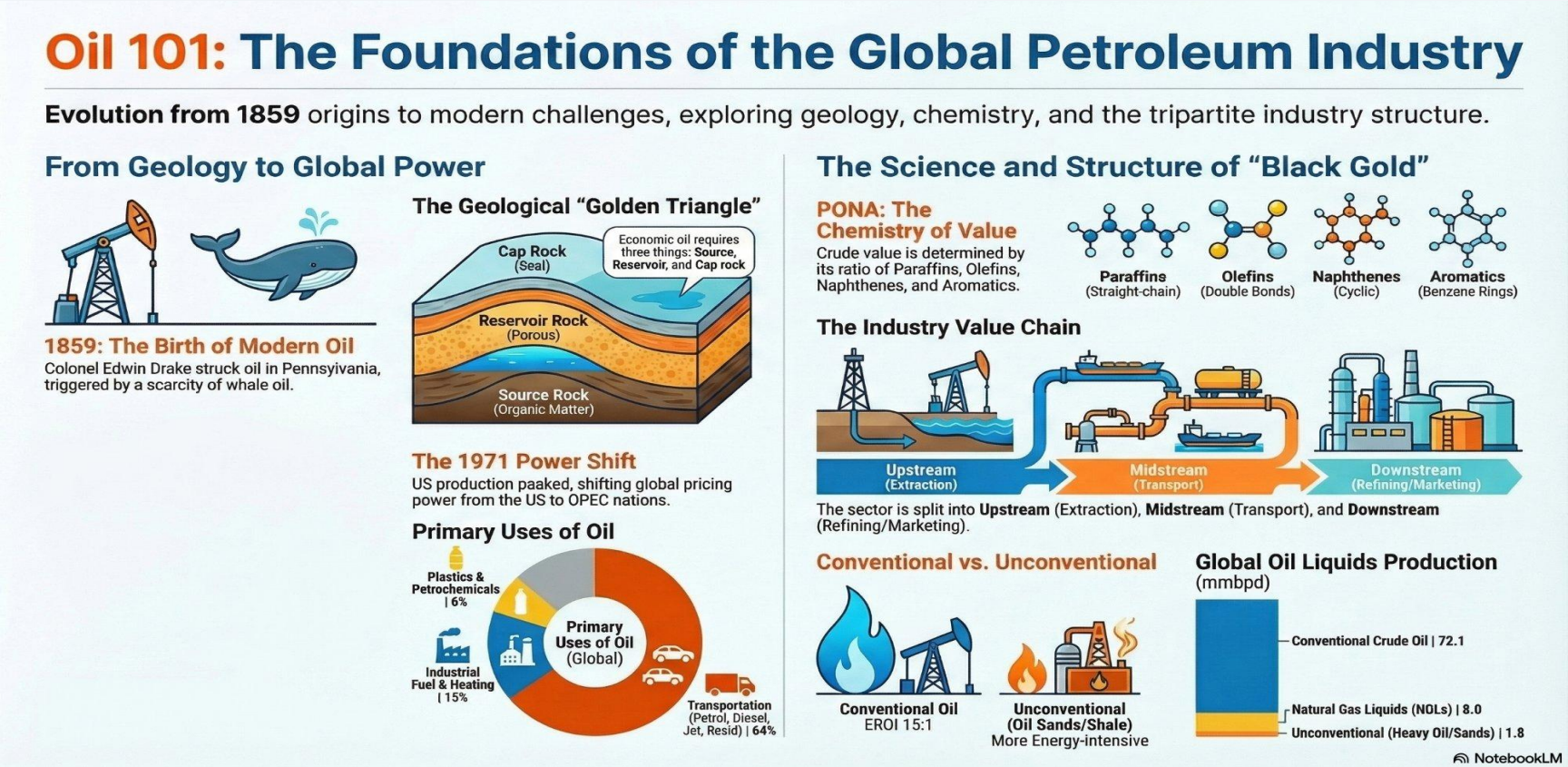
Beginner → Intermediate → Advanced guided learning journeys

Web Deployment

Interactive Gradio UI deployed to Render for global access

Current Data Source

Oil 101 by Morgan Downey — The Foundations of the Global Petroleum Industry



Technical Architecture

End-to-end system design: from PDF ingestion to web-deployed answers

PDF Ingestion
PyPDFLoader
+ metadata

›

Chunking & Embedding
500-char chunks
MiniLM-L6-v2

›

Vector Store
ChromaDB
with citations

›

LangGraph Router
Gemini / Qwen
fallback

›

Gradio UI
Deployed on
Render

Data Source

Oil 101 by Morgan Downey (PDF)
Stored on Google Drive

Secret Keys

GEMINI_API_KEY
HF_TOKEN

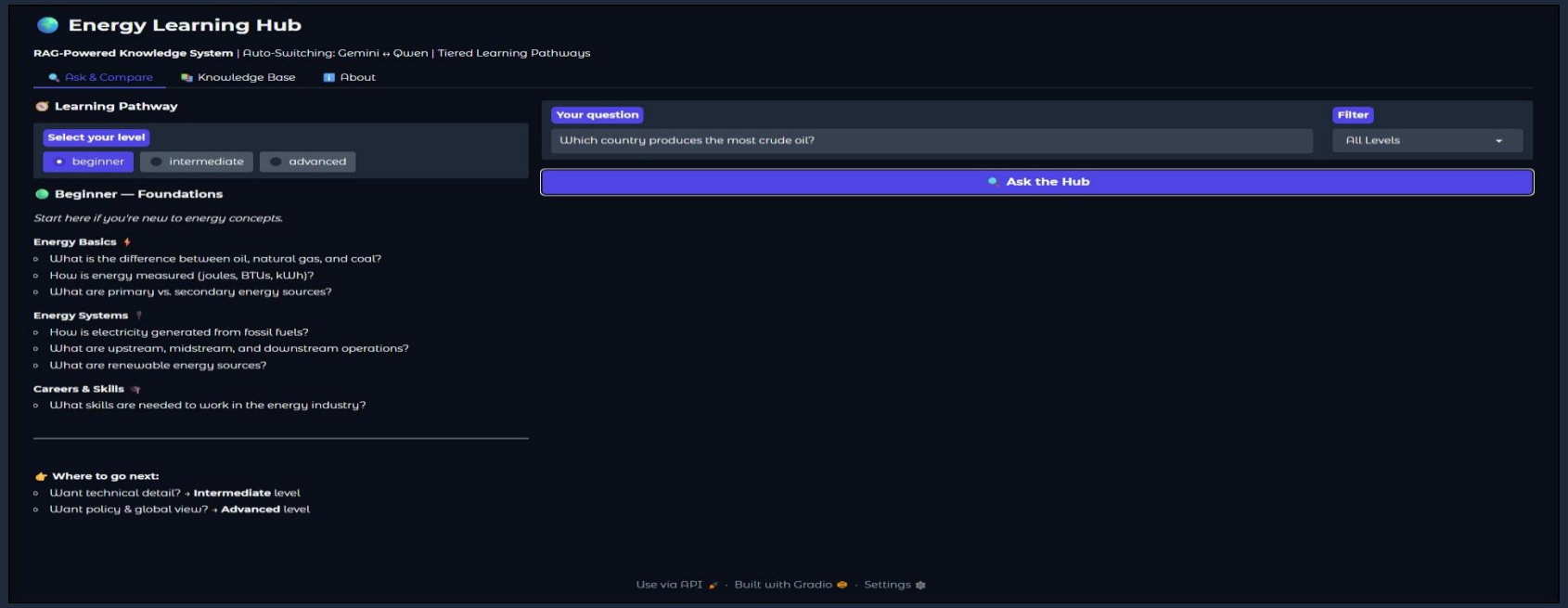
Output

Cited RAG answers + side-by-side
LLM comparison for learners

Live on the Web

The Energy Learning Hub is deployed and accessible worldwide

https://energylearninghub.onrender.com



LLM Strategy

Two models orchestrated by LangGraph for resilient, comparable responses

Google Gemini

Primary LLM

`gemini-2.5-flash`

Fast, cost-effective general reasoning. Handles the majority of learner queries with low latency.

`Key: GEMINI_API_KEY`

Qwen (HuggingFace)

Fallback LLM

`Qwen2.5-7B-Instruct`

Open-source fallback via HuggingFace Inference API. Activated when Gemini hits its quota limit.

`Key: HF_TOKEN`

LangGraph Router: Fan-out retrieval → parallel LLM calls → merge results → display with model attribution

Future Plans

Scaling the hub beyond Oil 101 into a comprehensive energy learning platform

Mobile Platform

Responsive mobile-first UI so learners can study on-the-go. Explore native iOS/Android app with offline cached FAQs and push notifications for new content.

Multiple Data Sources

Ingest additional PDFs, IEA/OPEC reports, academic journals, and industry whitepapers. Support Word docs, web scraping, and structured data alongside PDFs.

Additional LLM Providers

Integrate Claude, Llama, and other LLMs via the LangGraph router. Enable learners to compare answers across even more models for richer insights.

Assessments & Progress

Add micro-quizzes at the end of each tier. Track learner progress with completion badges and personalized recommendations for next topics.

Summary & Next Steps

- RAG system grounded in Oil 101 by Morgan Downey delivers cited, accurate answers
- LangGraph orchestrates Gemini and Qwen with automatic failover
- Tiered FAQ pathways (21 questions) guide learners from basics to strategic depth
- Live at energylearninghub.onrender.com — interactive, always-on web access

Next Steps

- Launch mobile-responsive version and explore native mobile apps
- Ingest IEA, OPEC, and additional textbook PDFs to expand knowledge base
- Integrate additional LLM providers (Claude, Llama) for broader comparison
- Add learner assessments, progress tracking, and completion badges